Laws of Indices

Exam code: 0607

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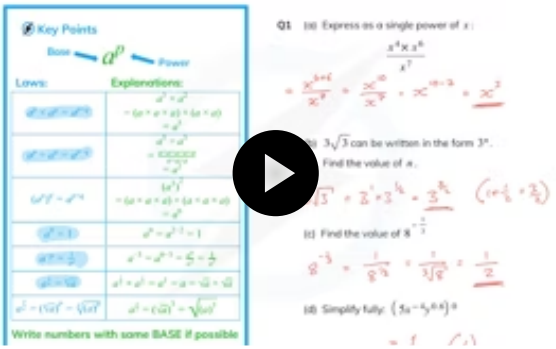
Written by: Jamie Wood
Reviewed by: Dan Finlay
Updated on 28 January 2026



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Laws of indices

What are the laws of indices?

- **Index laws** are **rules** you can use when doing **operations** with **powers**
  - They work with both **numbers** and **algebra**

Law	Description	How it works
$a^1 = a$	Anything to the power of 1 is itself	$6^1 = 6$
$a^0 = 1$	Anything to the power of 0 is 1	$8^0 = 1$
$a^m \times a^n = a^{m+n}$	To multiply indices with the same base, add their powers	$4^3 \times 4^2$ $= (4 \times 4 \times 4) \times (4 \times 4)$ $= 4^5$

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Got it

	the same base, subtract their powers	7×7 $= 7^3$
$(a^m)^n = a^{mn}$	To raise indices to a new power, multiply their powers	$(14^3)^2$ $= (14 \times 14 \times 14) \times (14 \times$ $= 14^6$
$(ab)^n = a^n b^n$	To raise a product to a power, apply the power to both numbers, and multiply	$(3 \times 4)^2 = 3^2 \times 4^2$
$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	To raise a fraction to a power, apply the power to both the numerator and denominator	$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2} = \frac{9}{16}$
$a^{-1} = \frac{1}{a}$ $a^{-n} = \frac{1}{a^n}$	A negative power is the reciprocal	$6^{-1} = \frac{1}{6}$ $11^{-3} = \frac{1}{11^3}$
$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$	A fraction to a negative power, is the reciprocal of the fraction, to the positive power	$\left(\frac{2}{5}\right)^{-3} = \left(\frac{5}{2}\right)^3 = \frac{5^3}{2^3} =$
$a^{\frac{1}{n}} = \sqrt[n]{a}$	A number raised to the fractional power $\frac{1}{n}$ is the same as the nth root if the number	$25^{\frac{1}{2}} = \sqrt[2]{25} = 5$ $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$
$a^{-\frac{1}{n}} = \frac{1}{a^{\frac{1}{n}}} = \frac{1}{\sqrt[n]{a}}$	A negative, fractional power is one over a root	$64^{-\frac{1}{2}} = \frac{1}{\sqrt{64}} = \frac{1}{8}$ $125^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{125}} = \frac{1}{5}$

Help me understand

$= (a^n)^{\ } = (a^m)^n$	power $\frac{\ }{n}$ is the nth root all to the power m, $(\sqrt[n]{\ })^m$, or the nth root of the power m, $\sqrt[n]{(\ )^m}$ (both are the same)	$8^{\frac{2}{3}} = (8^2)^{\frac{1}{3}} = \sqrt[3]{64} = 4$

How do I deal with different bases?

- Index laws only work with terms that have the **same base**
  - $2^3 \times 5^2$ cannot be simplified using index laws
- Sometimes expressions involve different base values, but **one is related to the other by a power**
  - e.g. $2^5 \times 4^3$
- You can use powers to **rewrite one of the bases**
  - $2^5 \times \mathbf{4}^3 = 2^5 \times (\mathbf{2}^2)^3$
  - This can then be simplified more easily, as the two bases are now the same
  - $2^5 \times (2^2)^3 = 2^5 \times 2^6 = 2^{11}$

Worked Example

(a) Find the value of x when $6^{10} \times 6^x = 6^2$

Answer:

Using the law of indices $a^m \times a^n = a^{m+n}$ we can rewrite the left hand side

$$6^{10} \times 6^x = 6^{10+x}$$

So the equation is now

$$6^{10+x} = 6^2$$

Comparing both sides, the bases are the same, so we can say that

$$10 + x = 2$$

Subtract 10 from both sides

$$x = -8$$

(b) Find the value of n when $5^n \div 5^4 = 5^6$

Answer:

Using the law of indices $a^m \div a^n = a^{m-n}$ we can rewrite the left hand side

$$5^n \div 5^4 = 5^{n-4}$$

Comparing both sides, the bases are the same, so we can say that

$$n - 4 = 6$$

Add 4 to both sides

$$n = 10$$

(c) Without using a calculator, find the value of 2^{-4}

Answer:

Using the law of indices $a^{-n} = \frac{1}{a^n}$ we can rewrite the expression

$$2^{-4} = \frac{1}{2^4}$$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$ so we can rewrite the expression

$$\frac{1}{2^4} = \frac{1}{16}$$

$$\frac{1}{16}$$

(d) Without using a calculator, find the value of $8^{-\frac{1}{3}}$

Answer:

Using the law of indices $a^{-n} = \frac{1}{a^n}$ we can rewrite the expression

$$8^{-\frac{1}{3}} = \frac{1}{8^{\frac{1}{3}}}$$

Using the law of indices $a^{\frac{1}{n}} = \sqrt[n]{a}$ we can rewrite the expression

$$\frac{1}{8^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{8}}$$

The cube root of 8 is 2

$$\frac{1}{2}$$

(e) Without using a calculator, find the value of $81^{\frac{3}{4}}$.

Answer:

Use the law of indices $(a^m)^n = a^{mn}$ we can rewrite the expression in two ways

$$81^{\frac{3}{4}} = (81^3)^{\frac{1}{4}} \text{ or } \left(81^{\frac{1}{4}}\right)^3$$

Both forms are equivalent, but $(81^3)^{\frac{1}{4}}$ would require calculating 81 cubed, so use the second form instead

$$\left(81^{\frac{1}{4}}\right)^3 = \left(\sqrt[4]{81}\right)^3$$

The 4th root of 81 is 3 as $3 \times 3 \times 3 \times 3 = 3^4 = 81$

$$(3)^3$$

Lastly, calculate or recall 3 cubed

$$27$$

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